Program : **Diploma in Chemical Engineering**	
Course Code : **5073B**	Course Title: **Modern Separation Techniques**
Semester : **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L: 4 T: 0 P: 0)**	Periods per semester: **60**

Course Objectives:

- To enable the students to comprehend the principles of modern separation techniques and its application in Industries.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Diffusion mechanism, Adsorption operations		Mass transfer operations I	4

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the principles and mechanism of adsorption, centrifugal separation and chromatography	12	Understanding
CO2	Summarize membrane separation processes.	18	Understanding
CO3	Summarize charge based separation techniques	18	Understanding
CO4	Summarise liquid membranes and surfactant based separations	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Summarize the principles and mechanism of adsorption, centrifugal separation and chromatography**		
M1.01	Summarize the difference between conventional and novel separation techniques	3	Understanding
M1.02	Summarise the working of various adsorption equipments	3	Understanding
M1.03	Study the various types of centrifugal separation techniques	3	Understanding
M1.04	Illustrate the principles and types of chromatographic techniques	3	Understanding
Contents: Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications.			
CO2	**Summarize membrane separation processes.**		
M2.01	Summarise the principle of membrane separations process	4	Understanding
M2.02	Study the various types of synthetic membranes	5	Understanding

<table>
<tr><td>M2.03</td><td>Illustrate the various membrane modules and applications</td><td>5</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Summarise the methods of membrane manufacture</td><td>4</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td></td><td></td></tr>
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Contents:

Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning.

Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane.

Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module.

General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes.

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<tr><td>CO3</td><td>Summarize charge based separation techniques</td><td></td><td></td></tr>
<tr><td>M3.01</td><td>Summarise the reverse osmosis process</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Summarise the filtration processes and their applications</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Outline electrodialysis and pervaporation</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Outline the technique of gas separation using membranes</td><td>4</td><td>Understanding</td></tr>
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Contents:

Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications.

Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications.

Nanofiltration- principle, NF membranes, Process limitations, Industrial applications.

Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications.

Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis.

Gas separation using membranes- cross flow and counter current model.

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<tr><td>CO4</td><td colspan="3">Summarise liquid membranes and surfactant based separations</td></tr>
<tr><td>M4.01</td><td>Summarise the technique of liquid separation using membranes</td><td>4</td><td>Understanding</td></tr>
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M4.02	Summarise the methods for foam and bubble separation	2	Understanding
M4.03	Summarise froth flotation and foam fractionation	2	Understanding
M4.04	Summarise the concept of Cryogenic fractionation	4	Understanding
	Series Test – II		

Contents:

Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes.

Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation.

Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation.

Cryogenic fractionation- concept and applications.

Text / Reference

T/R	Book Title/Author
T1	New Chemical Engineering Separation Techniques ; Schoen H. M
R1	Separation ‘Process Principles; Seader J.D, and Henley E.J
R2	Separation Processes; King C.J
R3	Perry’s Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W
R4	Handbook of Separation Process Technology; Rousseu, R.W

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/105/103105060